Joshua Rosenthal

+31 629 774 775 · joshuarosenthalza@gmail.com · LinkedIn · GitHub · Scholar

Summary

MSc Artificial Intelligence candidate (University of Amsterdam), first-class BSc in Computer Science and Applied Statistics, with research and engineering experience spanning generative modelling, production ML systems, and client-facing solutions architecture. My work focuses on deep generative modelling with an emphasis on efficiency, scalability, and practical deployment. Eligible to work in the EU.

Publications

Crystalite: A Lightweight Transformer for Efficient Crystal Modeling *2026*

Tin Hadži Veljković*, **Joshua Rosenthal***, Ivor Lončarić, Jan-Willem van de Meent

ICML AI4Physics · under review, NeurIPS '26 [Code] [Paper]

Symmetry-Aware Graph Metanetwork Autoencoders: Model Merging through Parameter Canonicalization *2025*

Odysseas Boufalis*, Jorge Carrasco-Pollo*, **Joshua Rosenthal***, Eduardo Terrés-Caballero*, Alejandro García-Castellanos

TAG-DS Conference · Spotlight · PMLR [Paper]

Efficient Optimization of Hierarchical Identifiers for Generative Recommendation *2025*

Federica Valeau*, Odysseas Boufalis*, Polytimi Gkotsi*, **Joshua Rosenthal***, David Vos

ECIR · Springer LNCS [Code] [Paper]

[Re] Benchmarking LLM Capabilities in Negotiation through Scoreable Games *2025*

Jorge Carrasco Pollo*, Ioannis Kapetangeorgis*, **Joshua Rosenthal***, John Hua Yao*

TMLR [Code] [Paper]

** Equal contribution.*

Experience

Research Intern & MSc Thesis **Amsterdam, NL**

UvA-Bosch Delta Lab *Jan 2026 – Jun 2026*

- Developed **Crystalite**: a lightweight diffusion transformer, implementing the core architecture and optimizing inference pipelines to sample 10–100× faster than equivariant baselines.
- Engineered **Flywheel**, an MLIP-based self-improvement loop that raised metastable-novel crystal discovery from 22.6% to 35% over two iterative training rounds. [Thesis Code]

Research Engineer (Volunteer) **Amsterdam, NL**

Dutch NAO Team (Whirlwind), Intelligent Robotics Lab, UvA *Nov 2024 – Nov 2025*

- Designed and built a reinforcement learning simulation framework in MuJoCo for autonomous robotic soccer.
- Implemented, trained, and evaluated single- and multi-agent reinforcement learning policies (PPO, MAPPO).

Data Engineer **Remote**

CallCabinet *May 2024 – Jul 2024*

- Fine-tuned a BERT model for emotion and topic classification on a large corpus of call-centre transcripts.
- Designed and built a production retrieval-augmented generation (RAG) pipeline (Python, GPT-3.5, Azure): embedded large-scale customer-interaction data and indexed it in Elasticsearch for semantic search.

Solutions Architect **Cape Town, ZA**

GoSolr *Feb 2023 – May 2024*

- Built and shipped REST APIs and backend services on AWS (TypeScript, SQL), spanning high-level API infrastructure to distributed warehouse-management integration.
- Architected distributed energy-management systems and inverter communication protocols for smart-home deployments serving 10,000+ consumers.

Education

University of Amsterdam **Amsterdam, NL**

MSc Artificial Intelligence *Sep 2024 – Jun 2026 (Expected)*

- GPA **8.6 / 10**. Coursework: Deep Learning, Geometric Deep Learning, Reinforcement Learning, Causality, Information Retrieval.

University of Cape Town **Cape Town, ZA**

BSc (Honours) Computer Science and Applied Statistics *Feb 2019 – Nov 2022*

- **82.1%**, First Class Honours, Dean's Merit List 2019–2022. Focus: machine learning, optimisation, statistical inference.

Selected Projects

- **Causal Effect of Education on Wages.** Full causal-inference pipeline reproducing Card (1995): DAG specification, by-hand backdoor + IV identification, conditional-independence tests, and IV estimation. [Code]
- **Ray Tracer** (Rust). Built from scratch, working through the rendering and graphics math. [Code]

Technical Skills

Languages: Python, Rust, TypeScript, SQL, Java

ML & Scientific Python: PyTorch, TensorFlow, Hugging Face, NumPy, SciPy, Pandas, pymatgen

Infrastructure & Tooling: AWS, Azure, SLURM, Weights & Biases, Git

Databases & Search: PostgreSQL, MongoDB, Elasticsearch